

Static Cosmology

Lecture 8: Midterm Data Review

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Overview

The midterm review checks the common calibration and uncertainty model used by the first seven data sets.

1 Observational setup

Source coordinates, observing band, selection function, and calibration state are fixed at the outset.

$$manifest = hash(catalog, instruments, CMT, atlas)$$

2 Transport or equilibrium equation

The governing equation is solved first in a homogeneous limit and then with the stated spatial dependence.

$$L_{mid} = L_z + L_T + L_Y + L_d + L_\rho$$

3 Connection to data

Each model quantity is tied to a measured flux, spectrum, angle, count, or temperature.

$$check = ETHERMAP \wedge STASIS \wedge human$$

4 Degeneracies

Calibration, population, and medium parameters are varied separately to expose their degeneracies.

5 Example

The worked example carries units and uncertainty from the input catalog to the reported result.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Weeks 1–7 notes
- Catalog Review Form
- CMT-9 checksum list